Nitin Panchal

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portfolio-nitinpanchal19.vercel.app

Experience

M37 Labs Gurugram, Haryana, India
Full Stack AI Engineer *October 2025 – June 2026*

- Architected and delivered an intelligent **risk-monitoring dashboard** for public-relations firms, owning end-to-end system design, backend development, and AI integration — **serving 10+ client orgs and 600+ users in production**.
- Designed **multi-tenant architecture** with strict tenant isolation, scoped **RBAC**, policy-driven permissions, and configurable organization hierarchies with automated isolation testing, enforcing zero cross-tenant data access.
- Built a **model-orchestration layer** routing inference across 4 LLM providers (OpenAI, Anthropic, Google, and local open-source models) based on task type, latency SLA, and cost constraints — engineered to cut inference spend 40% by tiering requests, with **local models** handling **privacy-sensitive workloads** on-prem.
- Developed FastAPI microservices and Celery-based distributed task queues powering a unified **voice + chat agent** with multi-turn memory, tool calling, and streaming STT/LLM/TTS — designed for sub-second (∼800ms) first-response latency and **50+ concurrent sessions**.
- Own the backend of a clienteling application for a leading Indian jewellery brand, designing and shipping FastAPI services and managing **Azure** deployments via **Jenkins CI/CD**.

AventIQ Delhi, India
Software Developer *November 2024 – October 2025*

- Built a full-stack MERN Employee Management System with modules for onboarding, attendance, leave, asset tracking, and task management, integrating **Microsoft Graph APIs** to auto-provision employees in **Azure AD**.
- Developed **role-based dashboards** for employees and HR using React and Tailwind CSS, enabling real-time insights into attendance, performance, and pending tasks.
- Integrated **Razorpay** payments and **AWS Cognito** authentication into a **serverless** senior-citizen healthcare application, **reducing server maintenance costs by 30%**.
- Developed a React-based admin dashboard on AWS with IAM RBAC to manage student records and university appointments, optimizing MySQL stored procedures for a **20% query performance gain**.

Projects

Themis: Autonomous Compliance Agent
Next.js · FastAPI · RAG · Qdrant · PostgreSQL · Redis · Celery · Docker · LiteLLM

- Built a secure multi-tenant agentic SaaS with a 3-stage pipeline (policy RAG → LLM reasoning → deterministic risk scoring) that ingests 5 document formats and detects citation-backed violations across GDPR, HIPAA, PCI-DSS, and SOX, **cutting manual review effort ∼80%**.
- Built a RAG engine on Qdrant with 1536-dim OpenAI embeddings, retrieving the top-5 policy clauses per chunk to ground the LLM and eliminate hallucinated citations, **lifting detection precision ∼35%**.
- Engineered an async Celery pipeline (S3 → parse → chunk → embed → index) with retries, **sustaining ∼99% task reliability** on 50 MB+ uploads and feeding a confidence-weighted 0–100 risk engine; enforced tenant isolation, 3-role RBAC, JWT auth (15-min access + rotating 7-day refresh), rate limits, request-size caps, and security headers.

AgentScope: AI Agent Observability & Evaluation Platform
Next.js · FastAPI · PostgreSQL · Redis · Celery · Qdrant · Docker

- Built a multi-tenant platform for **tracing, evaluating, replaying, and regression-testing AI agents**, with hierarchical spans, token/cost attribution, live updates, and tenant-scoped analytics.
- Designed an **idempotent telemetry pipeline** using a fault-tolerant Python SDK and transactional FastAPI ingestion, with sampling, batching, redaction, retry/backoff, rotating tokens, scoped API keys, RBAC, and cross-tenant isolation.
- Developed a distributed evaluation and experimentation system with deterministic, semantic, and LLM-as-judge evaluators, immutable datasets, isolated Celery queues, SSRF-protected execution, baseline comparisons, **CI quality gates**, alerting, and retention.

Skills

AI/ML	RAG, Agentic Workflows, LLM Fine-tuning, OpenAI Agents SDK, LiteLLM, Vector Databases (Qdrant), Ollama, Hugging Face
Backend	FastAPI, Node.js, Express.js, RESTful APIs, Celery, Redis
Languages	Python, JavaScript, TypeScript, C++, C
Frontend	React.js, Next.js, Tailwind CSS, HTML5, CSS3, Redux
Databases	PostgreSQL, MongoDB, MySQL
Cloud	AWS (Cognito, API Gateway, Lambda, S3, EC2, IAM, RDS, SES, CloudWatch, Secrets Manager, SQS, CloudFront), Azure
Tools	Docker, Git, GitHub, Jenkins, Postman, Vercel, Netlify

Education

Guru Gobind Singh Indraprastha University Delhi, India
B.Tech (Computer Science and Engineering) — CGPA: 9.1/10 *September 2020 – June 2024*